

Time, Computation, and Novelty: A No-Go Theorem from Petz Recovery

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We prove a no-go theorem connecting time-reversibility, quantum computation, and the arrow of time within the Petz recovery framework. A quantum channel has $\tau = 0$ for all input states if and only if it is unitary (Theorem 1). It follows that ideal unitary circuits are simultaneously time-reversible and computationally universal (Theorem 2), while any physical gate with noise breaks time symmetry (Theorem 3). These results yield a trilemma: a physical system can have at most two of {time-reversibility, quantum coherence, environmental interaction} (Corollary 4). The saturation surface of the Petz recovery bound is observer-dependent—the “boundary of the time machine” depends on the observer’s prior (Theorem 5). For non-Markovian dynamics, local time reversal ($d\tau/dt < 0$) is possible but fundamentally bounded by envelope decay and total entropy production. The past has not disappeared; it is information leaked to the environment. τ measures how much has leaked. Time is the cost of novelty.

INTRODUCTION

The quantum eraser [1, 2] demonstrates that a future measurement on an idler photon can restore interference in a signal photon whose which-path information has been recorded. The future measurement affects the present interpretation. But there is an asymmetry: in an open system, one must wait for the future measurement to actually occur before the past can be reinterpreted. The arrow of time appears precisely because the system is open.

The Petz recovery map [3, 4] provides the mathematical structure underlying this observation. Given a quantum channel $\mathcal{N}$ and a reference state σ , the Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ is the unique Bayesian retrodiction functor [5]: it recovers the past from the future. The recovery infidelity $\tau = 1 - F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ measures how much the past differs from its retrodicted version—the operational arrow of time. When $\tau = 0$ for all inputs, prediction and retrodiction are equivalent, and time has no direction. When $\tau > 0$, some information about the past is gone, and the arrow of time exists.

In Ref. [6] (hereafter Paper 1), we established the equivalence chain $\tau = 0 \Leftrightarrow \Sigma = 0 \Leftrightarrow I(A; E|B) = 0 \Leftrightarrow$ QMC linking vanishing temporal asymmetry to vanishing entropy production, vanishing information leakage, and the quantum Markov condition. We also proved a composition theorem showing that $\sqrt{\tau}$ satisfies a triangle inequality under channel composition.

In this Letter, we use these tools to prove a no-go theorem for “time machines.” The central question is: can a physical system be simultaneously time-reversible ($\tau = 0$) and computationally useful? The answer reveals a trilemma connecting time-reversibility, quantum coherence, and environmental interaction.

The physical picture that emerges is this: the past has not disappeared. It is encoded in the environment—in the photons that scattered off objects, the thermal vibrations of air molecules, the decoherence records in

the surrounding matter. The parameter τ measures how much information has leaked. A “time machine,” in this framework, is not a device that bends spacetime. It is a device that recovers information from the environment.

FRAMEWORK

We work within the framework of Paper 1 [6]. All Hilbert spaces are finite-dimensional.

The *temporal asymmetry parameter* for a CPTP map $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$, input state $\rho \in \mathcal{S}(\mathcal{H}_A)$, and faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$ is

$$\tau(\rho, \mathcal{N}, \sigma) = 1 - F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))), \quad (1)$$

where $\mathcal{R}_{\sigma,\mathcal{N}}$ is the Petz recovery map [3, 4] and F is the Uhlmann fidelity.

A CPTP map $\mathcal{N}$ is *time-reversible* if $\tau(\rho, \mathcal{N}, \sigma) = 0$ for all states ρ and all faithful σ . Equivalently (Paper 1, Eq. (10)): $\tau = 0$ for all inputs iff $\Sigma = 0$ iff the quantum Markov condition holds iff perfect Petz recovery is achieved.

A finite set of CPTP maps $\mathcal{G} = \{\mathcal{N}_1, \dots, \mathcal{N}_k\}$ acting on $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ is *computationally universal* if, for any unitary $U \in U(2^n)$ and any $\epsilon > 0$, some finite composition $\mathcal{N}_{i_m} \circ \dots \circ \mathcal{N}_{i_1}$ approximates $U\rho U^\dagger$ to within ϵ in trace norm for all ρ .

A CPTP map is *noiseless* if it is a unitary conjugation $\mathcal{N}(\rho) = V\rho V^\dagger$; otherwise it is *noisy*.

THE NO-GO THEOREM

Theorem 1 (Structure of time-reversible channels). *Let $\mathcal{N} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ be a CPTP map. Then $\tau(\rho, \mathcal{N}, \sigma) = 0$ for all states ρ and all faithful σ if and only if $\mathcal{N}$ is a unitary conjugation: $\mathcal{N}(\rho) = U\rho U^\dagger$ for some unitary U .*

Proof.—The “if” direction is Corollary 1 of Paper 1 [6]: for unitary $\mathcal{N}_U$, the Petz recovery is $\mathcal{R}_{\sigma, \mathcal{N}_U}(X) = U^\dagger X U$ independent of σ , giving $F = 1$ and $\tau = 0$.

For “only if”: $\tau = 0$ for all ρ means $F(\rho, \mathcal{R}_{\sigma, \mathcal{N}}(\mathcal{N}(\rho))) = 1$ for all ρ . By the Petz sufficiency theorem [4], this holds for all ρ if and only if $\mathcal{N}$ is sufficient for all pairs (ρ, σ) , which requires $\mathcal{N}$ to be reversible. A reversible CPTP map on a finite-dimensional algebra is a $*$ -automorphism, which by the Wigner theorem is unitarily implemented. $\square$

Theorem 2 (Ideal computation is time-reversible). *Let $\mathcal{G} = \{\mathcal{N}_1, \dots, \mathcal{N}_k\}$ be a gate set where each $\mathcal{N}_i$ is a unitary conjugation. Then:*

- (a) *Each $\mathcal{N}_i$ has $\tau_i = 0$.*
- (b) *Any composition $\mathcal{N}_{i_m} \circ \dots \circ \mathcal{N}_{i_1}$ is unitary, hence has $\tau_{\text{total}} = 0$.*
- (c) *If the unitaries $\{U_1, \dots, U_k\}$ form a universal gate set (e.g., $\{H, T, \text{CNOT}\}$), then $\mathcal{G}$ is computationally universal.*

Therefore, ideal quantum computation is simultaneously time-reversible and computationally universal.

Proof.—(a) and (b) follow from Corollary 1 of Paper 1 and the fact that the composition of unitaries is unitary. (c) is the Solovay-Kitaev theorem [7, 8]: any dense subgroup of $\text{SU}(d)$ provides ϵ -approximate universality with gate count $O(\log^c(1/\epsilon))$. $\square$

Theorem 3 (Noisy gates break time symmetry). *Let $\mathcal{N}$ be a CPTP map that is not a unitary conjugation. Then there exist states ρ and σ such that $\tau(\rho, \mathcal{N}, \sigma) > 0$.*

Proof.—Contrapositive of Theorem 1. $\square$

The naive argument and its failure.—One might attempt a different proof: from the saturation theorem (Paper 1, Theorem 4), $\tau = 0$ requires $[\mathcal{N}(\rho), \mathcal{N}(\sigma)] = 0$, so a time-reversible channel maps everything to a commuting subalgebra and cannot generate quantum coherence. This argument has a gap. The saturation theorem characterizes when $F^2 = \exp(-\Delta D)$ is an equality—but $\tau = 0$ requires $F = 1$, a stronger condition. The correct route is through Theorem 1, which uses the full Petz sufficiency theorem rather than the saturation condition.

The key point is that Theorem 2 appears to “destroy” the no-go theorem: ideal quantum computation has $\tau = 0$. The real no-go emerges from the distinction between mathematical and physical computation. Any physical quantum computer operates with noisy gates. The physical process implementing gate U_i is not the ideal unitary conjugation but a noisy CPTP map that approximates it. By Theorem 3, every such gate contributes $\tau > 0$.

The arrow of time in a quantum computer comes from noise, not from computation. The unitary content of

computation contributes zero to the time arrow. Only the irreversible periphery—initialization, measurement, decoherence—has $\tau > 0$.

THE TRILEMMA

Corollary 4 (The time-coherence-environment trilemma). *A physical system can achieve at most two of the following three properties simultaneously:*

- (i) **Time-reversibility:** $\tau = 0$ for all inputs.
- (ii) **Quantum coherence:** non-commuting output states.
- (iii) **Environmental interaction:** $\mathcal{N}$ is not unitary.

Any two imply the negation of the third.

Proof.—Three cases.

(i) + (ii) $\Rightarrow \neg$ (iii): Time-reversible quantum coherence (non-commuting outputs with $\tau = 0$ for all inputs) requires, by Theorem 1, that $\mathcal{N}$ is unitary—no environmental interaction.

(i) + (iii) $\Rightarrow \neg$ (ii): A time-reversible non-unitary channel has $\tau = 0$ for all inputs and is non-unitary. But Theorem 1 says $\tau = 0$ for all inputs requires unitarity—contradiction unless we weaken to state-specific $\tau = 0$. A non-unitary channel with $\tau(\rho, \mathcal{N}, \sigma) = 0$ for specific pairs maps those states to a commuting subalgebra (Paper 1, Theorem 4), so there is no quantum coherence between them.

(ii) + (iii) $\Rightarrow \neg$ (i): Quantum coherence in an open system means $\mathcal{N}$ is not unitary and produces non-commuting outputs. By Theorem 3, $\tau > 0$ for some inputs. $\square$

Table I summarizes the three scenarios.

TABLE I. The trilemma. Each row satisfies two of the three properties and violates the third. The column “Physical?” indicates whether the scenario is realizable in a laboratory.

Scenario	(i)	(ii)	(iii)	Physical?
Ideal quantum computer	✓	✓	×	No (requires zero noise)
Classical open system	✓	×	✓	Yes (commuting outputs)
Real quantum computer	×	✓	✓	Yes (arrow of time)

Connection to Landauer-Bennett.—Landauer [9] showed that erasure costs at least $k_B T \ln 2$ per bit. Bennett [10] showed that logically reversible computation can in principle be performed without erasure, at zero energy cost. The trilemma is the quantum-information-theoretic counterpart: unitary computation has $\tau = 0$; only the irreversible periphery (noise, measurement, erasure) has $\tau > 0$. The logical content of computation is free. The physical overhead is what costs.

OBSERVER-DEPENDENT BOUNDARY

The saturation surface of the Petz recovery bound (Paper 1, Theorem 4) divides the space of triples $(\mathcal{N}, \rho, \sigma)$ into a recoverable region ($\tau \approx 0$) and an irreversible region ($\tau \gg 0$).

Theorem 5 (Observer-dependent time-reversibility). *For the same channel $\mathcal{N}$ and the same input state ρ , two observers using different reference states $\sigma_1 \neq \sigma_2$ can disagree on whether the process is time-reversible:*

$$\tau(\rho, \mathcal{N}, \sigma_1) = 0 \not\Rightarrow \tau(\rho, \mathcal{N}, \sigma_2) = 0. \quad (2)$$

Proof.—By example. Consider the amplitude damping channel $\mathcal{N}_\gamma$ with $\gamma = 0.1$ and $\rho = |0\rangle\langle 0|$. Take $\sigma_1 = I/2$. Then $\mathcal{N}(\rho) = |0\rangle\langle 0|$ and $\mathcal{N}(\sigma_1) = \text{diag}((1+\gamma)/2, (1-\gamma)/2)$. Since $\mathcal{N}(\rho)$ is rank-1 and $[\mathcal{N}(\rho), \mathcal{N}(\sigma_1)] = 0$, the likelihood ratio is constant on the support, so the saturation conditions of Paper 1, Theorem 4, are met and τ is small.

Now take $\sigma_2 = |1\rangle\langle 1|$. Then $\mathcal{N}(\sigma_2) = \text{diag}(\gamma, 1-\gamma)$, giving a different likelihood ratio and a different relative entropy gap. The recovery fidelity changes, and in general $\tau(\rho, \mathcal{N}, \sigma_1) \neq \tau(\rho, \mathcal{N}, \sigma_2)$. $\square$

The “boundary of the time machine” is not absolute. It depends on what the observer considers the prior state of the world. An observer with a prior σ close to ρ (more information) sees a larger region of time-reversibility. An observer with less information sees a smaller region. This connects to the core result of Paper 1: the arrow of time is not an absolute property of physics but emerges from the relationship between the system and the observer’s prior.

In the Bures geometry, $\tau = 1 - F$ measures the Bures distance between ρ and its Petz-recovered version. The saturation surface can be visualized as a “cone” in Bures space: processes inside the cone are recoverable, processes outside are irreversible. The reference state σ determines the orientation of this cone.

NON-MARKOVIAN TIME REVERSAL

For Markovian dynamics (Lindbladian evolution), the temporal asymmetry satisfies [6]

$$\frac{d\tau}{dt} \sim \frac{\dot{\Sigma}}{2}(1 - \tau) \geq 0, \quad (3)$$

where $\dot{\Sigma} \geq 0$ is the instantaneous entropy production rate. The arrow of time only advances.

For non-Markovian dynamics, the divisibility condition fails [19] and $\dot{\Sigma}$ can be temporarily negative, opening the possibility of

$$\frac{d\tau}{dt} < 0 \quad (\text{local time reversal}). \quad (4)$$

Information flows back from the environment to the system, and the Petz recovery temporarily improves without intervention.

Jaynes-Cummings model.—A qubit coupled to a single cavity mode with Lorentzian spectral density $J(\omega) = \gamma_0 \lambda^2 / [2\pi((\omega_0 - \omega)^2 + \lambda^2)]$ provides a concrete example [17]. The decoherence function is

$$G(t) = e^{-\lambda t/2} \left[\cosh \frac{dt}{2} + \frac{\lambda}{d} \sinh \frac{dt}{2} \right], \quad (5)$$

where $d = \sqrt{\lambda^2 - 2\gamma_0\lambda}$.

In the weak-coupling regime ($\gamma_0/\lambda < 1/2$), $G(t)$ decays monotonically and $\tau(t)$ increases monotonically—standard Markovian behavior. In the strong-coupling regime ($\gamma_0/\lambda > 1/2$), d becomes imaginary, $G(t)$ oscillates, and $\tau(t)$ exhibits intervals where $d\tau/dt < 0$. The fidelity $F(t)$ temporarily increases: the Petz recovery improves, constituting a concrete non-Markovianity witness distinct from the BLP trace-distance witness [17] and the RHP divisibility witness [18].

Design principles.—Four conditions favor local time reversal: (1) a structured environment (spectral peaks, finite bandwidth); (2) strong coupling ($\gamma_0/\lambda > 1/2$); (3) a finite-dimensional environment (information mirror, not information sink); and (4) coherent system–environment interaction (dephasing-type interactions do not produce backflow).

Fundamental limits.—Even with optimal non-Markovian engineering, three constraints apply.

Limit 1: Envelope decay. The oscillations in $\tau(t)$ occur within a decaying envelope: $\tau(t) \sim 1 - |G(t)|^2$ with $|G(t)| \leq e^{-\lambda t/2}$. Local time reversal is always temporary.

Limit 2: Total entropy production. Even when $d\tau/dt < 0$ in the subsystem, the total entropy production $\Sigma_{\text{total}} \geq 0$ for system plus environment. The subsystem’s temporary τ decrease is paid for by entropy increase elsewhere.

Limit 3: Isolation paradox. To achieve $\tau \rightarrow 0$ permanently requires $\lambda \rightarrow 0$: zero dissipation, complete isolation. But a completely isolated system undergoes unitary evolution, and by Theorem 2 can perform quantum computation internally. The problem is that one cannot program it (programming requires interaction: $\tau > 0$), cannot read out the result (measurement requires interaction: $\tau > 0$), and cannot change the computation (the evolution is fixed by the initial Hamiltonian). A time machine that cannot be programmed or read out is not a useful computer.

This closes the loop to the no-go theorem.

THE PAST AS LEAKED INFORMATION

The preceding results lead to a unified physical interpretation.

The past has not disappeared. It is encoded in the environment—in the photons that scattered off objects,

the thermal vibrations of air molecules, the decoherence records in surrounding matter. The parameter τ measures how much information has leaked from the system to the environment. A “time machine,” in the τ framework, is not a device that bends spacetime. It is a device that recovers information from the environment.

Consider a universe with $\tau = 0$ everywhere. All evolution is unitary. All processes are perfectly reversible. The Petz map achieves exact recovery everywhere. Past and future are operationally indistinguishable. No information is created or destroyed. Nothing happens that was not already determined by initial conditions.

Now consider a universe with $\tau > 0$. Non-unitary processes exist. Some information leaks irreversibly to the environment. The past is not perfectly recoverable from the future. Time has a direction. Measurement outcomes are genuinely random. New information can be created.

The $\tau = 0$ universe is a time machine—but it contains no novelty. The $\tau > 0$ universe has novelty—but it cannot reverse time.

This tension is quantitative. The arrow of time τ is bounded from below by the non-commutativity of the channel outputs (the “computational content” of the process) and bounded from above by the entropy production. It accumulates sub-additively across sequential processes via the composition theorem (Paper 1, Theorem 5). Time is not an all-or-nothing binary. It is a continuous parameter.

Time is the cost of novelty.

DISCUSSION

Trinity of computational bounds.—The no-go theorem takes its place alongside two other fundamental constraints on physical computation. The Margolus–Levitin bound [12] limits how *fast* one can compute: $t_{\min} = \pi\hbar/(2\Delta E)$. The Landauer bound [9] limits how *cheaply* one can compute: $W_{\text{diss}} \geq k_B T \ln 2$ per bit erased. The Petz– τ bound limits how *reversibly* one can compute: $\tau > 0$ for any noisy gate. Speed, cost, and reversibility together define the physical space of realizable computation.

Connection to $ER=EPR$.—Maldacena and Susskind [22] conjectured that entanglement is geometrically dual to wormholes. Susskind [23] and Brown et al. [24] connected the growth of the Einstein–Rosen bridge to the growth of quantum computational complexity. In our framework, τ grows as complexity grows: the wormhole elongates as information leaks irreversibly. A “time machine” ($\tau = 0$) corresponds to a static wormhole with no complexity growth—possible only for an eternal black hole in perfect thermal equilibrium. The no-go theorem in this language becomes: one cannot simultaneously have a growing wormhole (useful computation) and a static one (time-reversibility).

Connection to the quantum eraser.—The quantum eraser is the physical paradigm from which the entire τ framework originates. The eraser shows that a future measurement can restore past coherence—but only in a closed system, and only for the specific entangled state. The τ framework generalizes this: in a closed system ($\tau = 0$), retrodiction is exact and temporal order is irrelevant. In an open system ($\tau > 0$), one must wait for the future to actually happen. The no-go theorem adds: the price of temporal coexistence ($\tau = 0$) is that the system cannot produce genuinely new information. In a $\tau = 0$ world, everything is predetermined by initial conditions. There are no surprises.

Outlook.—The composition theorem (Paper 1, Theorem 5) and the design principles of Sec. suggest an engineering program: a “ τ -Chrono” engine that uses Bayesian composition, saturation scanning, and non-Markovian backflow optimization to design protocols for quantum memory extension, passive error correction, and precision tests of time symmetry.

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